Improving the team's wall

A Professional report on improving the goalkeeper

DISCOVER YOUR WORLD

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Applied Data Science and Artificial Intelligence

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1 Introduction

The world of football has gotten unbelievably big in recent years, cementing itself as the biggest sport in the world in every possible aspect, and by a wide margin at that. There is no other world sport that comes close or ever will come close to the reach and popularity football has achieved. Since the size of the sport has inflated so much, the importance of winning games is astronomical, especially at higher levels, since it also makes people a lot of money. Of course, the traditional ways to improve in playing performance that have stuck with the sport for many years such as physicality, fitness, skills, shooting, dribbling, game IQ, etc, have always been instrumental in the success of teams and those factors will never lose its importance, but as the sport has evolved, as it has gotten more popular and far too big to only uphold it by traditional standards, it has had to incorporate modern technological advancements, just like the world around it, not only football. Nowadays technology and data has become a huge part in football, often being the decider in which team is the better one. Now top clubs have huge data networks and collect massive amounts of data on their players in order to improve the performance of them. Success is everything in football and using player data in smart ways is the best way to achieve that.

Going off that, the business problem that I have found and will solve for the client is that Goalkeeper for NAC Breda, the client, need improving, and I will find the best statistics to focus on and which ones will give an increase in both performance and market value. This is only for Goalkeepers or sole goalkeeper since there is only one at NAC Breda from looking at the data. The reason for only focusing on this position is for simplicity, as I think it will bring better results if the focus isn't as broad and to give more importance on goalkeepers since I think that the GK is often overlooked compared to all the other positions. So, if there isn't anyone else doing it then at least there's me who is covering it.

To put all of this into one clearly understandable question, it would be along the lines of: "What player statistics does the NAC Goalkeeper have to improve on to perform at higher levels". We will explore this further throughout the report.

2 Exploratory Data Analysis

2.1 High level overview of the dataset:

The dataset that is being used for everything regarding this project is a big dataset with various data points such as Players, Teams, Age, Height and many different football statistics like goals, shots blocked and many more. There are various stats for every type of position, as an example, for defenders there would be shots blocked and for midfielders it would be number of accurate passes completed. There are also various teams, leagues and countries represented in this dataset. This is not surprising as this dataset has over 16,000 rows of data, each one being a different player, team, and country along with their respective stats according to their position. The dataset has not been sourced by me, instead it was provided to me by the University and the client NAC Breda, and as for the collection methods, they are not known to me at this moment. Most of the datatypes of the dataset are numerical types, since a lot of it is comprised of football statistics which are commonly represented as numerical values and then the rest are categorical or textual data types.

```
16535 rows × 114 columns
```

Figure 1 Number of rows and columns in dataset

```
Number of Numerical Variables: 105
Numerical Variables:
Index(['Age', 'Market value', 'Matches played', 'Minutes played', 'Goals',
       'xG', 'Assists', 'xA', 'Duels per 90', 'Duels won, %',
       ...
       'Prevented goals per 90', 'Back passes received as GK per 90',
       'Exits per 90', 'Aerial duels per 90.1', 'Free kicks per 90',
       'Direct free kicks per 90', 'Direct free kicks on target, %',
       'Corners per 90', 'Penalties taken', 'Penalty conversion, %'],
      dtype='object', length=105)
```

Figure 2 Numerical values in dataset

2.2 Steps taken to clean the dataset:

For the missing values that were present in my dataset, to get rid of them I used a simple imputer with the mean strategy and then applied that to the dataset, which fills the missing values in both numerical and categorical columns with their respective means.

```python
# Identifying numerical objects in my dataframe
num_variables = df.select_dtypes(include=['number']).columns
obj_variables = df.select_dtypes(include=['object']).columns

# Using the mean strategy to create a simple imputer and applying them to the dataframe
# This fills the missing values in numerical columns with their respective means
nums_imputer = SimpleImputer(strategy='mean')
df[num_variables] = nums_imputer.fit_transform(df[num_variables])

# Here I created a simpleimputer using the most frequent strategy
# This fills the missing values with the most frequent value in the column
obj_imputer = SimpleImputer(strategy='most_frequent')
df[obj_variables] = obj_imputer.fit_transform(df[obj_variables])

msno.bar(df)
```

Figure 3 Cleaning the dataset

I also checked for duplicates on my dataset, in which there were none. For certain parts of my final template, I excluded all non-numerical columns, so that for some models and calculations, they would work better, but that is only temporary. Among all of this I also used a standard scaler on my data, that was so that all of the features could be on a similar scale, preventing larger features to dominate simply based on their enlarged size. It also cut the executing time of some certain cell blocks and models by more than half.

```python
scaler = StandardScaler()
X_imputed_scaled = pd.DataFrame(scaler.fit_transform(X_imputed), columns=X.columns)
```

Figure 4 Scaling the features

2.3 Key findings

One interesting detail about this dataset is in the ‘Positions’ column. It is a categorical column which contains the positions of each player featured in the dataset. Some examples of football positions would be Left Winger (LW is how it would be written in the dataset) which is an attacking position. Or another one would be Center back (CB) which is a defensive position on the field. The way these positions are written is something like so: LW, LWB, CF. The positions individual like they are supposed to be, they have been grouped together which makes working with it a bit harder. Sure, a player’s position can sometimes change, but a player will always have their flagship position, the one they play most of the time and the one they excel at. Because of this fact I had to split them up. Using the str.split by the comma symbol and then exploding the 'Position' column, I was able to fix this and more easily use this column for EDA and other purposes.

```
# Split the position that are together like (AMF, LMF, RMF...)
df = df.assign(Position=df['Position'].str.split(', ')).explode('Position')
```

Figure 5 Splitting positions

2.4 Visual techniques:

Throughout the template I used a number of different visualizations and plots to explain and showcase my data. The most common type of plot that was used is the bar chart visualization. It's easy to understand and implement and you can gain and understand a lot from it, even if it is one of the simpler visualizations out there. It provided me with a better understanding of the entire dataset which in turn will make everything else I try to make, that will include more complicated charts and plots as well as tricky machine learning models, more accurate and generally everything will be found with more success in most aspects of the template. Less is more.

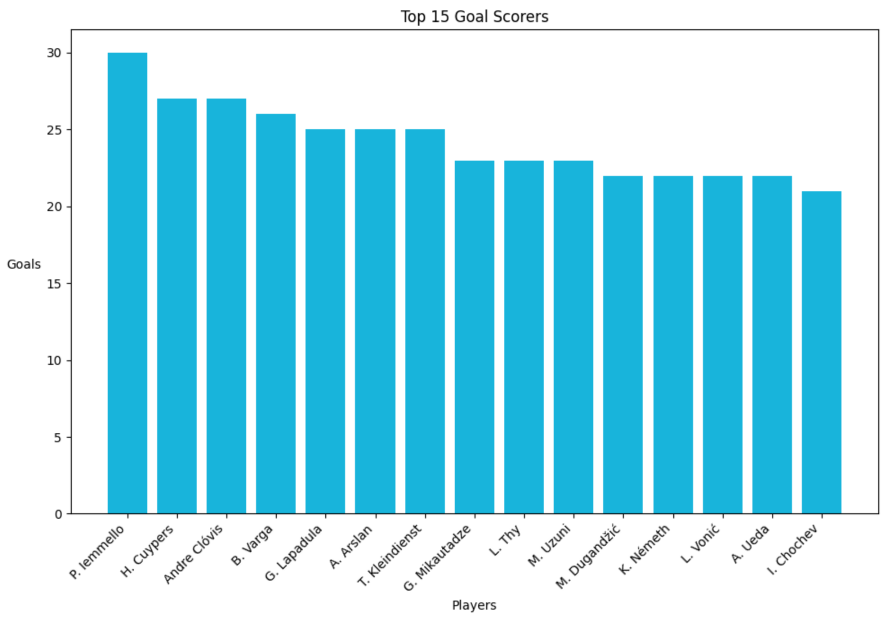

Figure 6 Top 15 goal scorers

Other types of visualizations include scatter plots which is very useful to analyse the strength between two variables and the outliers from the data. It especially helps with the linear regression model, which will be talked about a lot later. But pairing a scatter plot with the linear regression model can really show you the accuracy of the model you built or prediction that you are trying to make and will tell you many a thing that can be improved on and/or if what you built is adequate. They also help you make decisions on the data preprocessing, your feature selection, assumptions, and predictions, and it will overall contribute greatly to the final result. The insights I gained after using the scatter plot various times throughout the template, were that some of my models were really inaccurate. What that told me was that I of course had to change something about this. The data points were really dispersed and seldom data points made it near the line that indicates a perfect score. And the accuracy of the model was quite low, adding to the inaccuracy of the scatterplot. It told me which were the outliers and what I needed to do in order to improve it. I then fixed my models and plots and now the accuracy is very high and thanks to the original insights, I can have an accurate model providing even more and better insights than before.

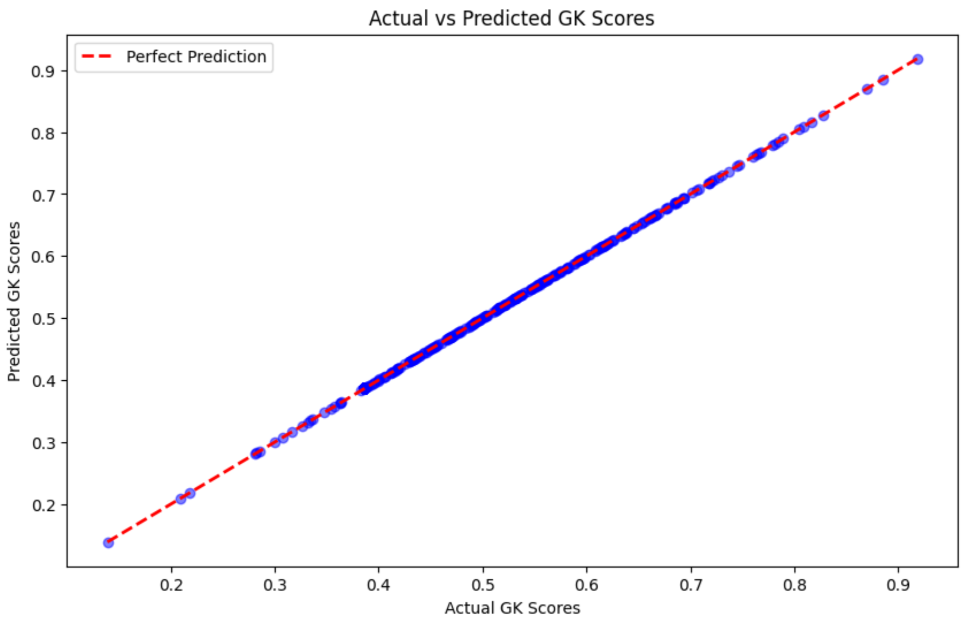

Figure 7 Actual vs Predicted GK Scores

The rest of the visualizations that I used to showcase some of the data include histograms and horizontal bar charts.

In the template I used Correlation Analysis to observe the relationship between some chosen features/variables which came with positive results. If there were to be an increase in a certain statistic then there would also be an increase in the other variable that was tested with it, and most of the time they scored above a 0.70 shown in the correlation matrix graph. As for my final model I used a standardized coefficient to gauge the relationship between my final chosen features that I used to predict the values I was trying to get. Standardized coefficients are particularly useful for comparing the importance of different predictors in a regression model, and it suits models like the linear regression, but I will touch more on that later. Overall, the relationships between the variables used are strong.

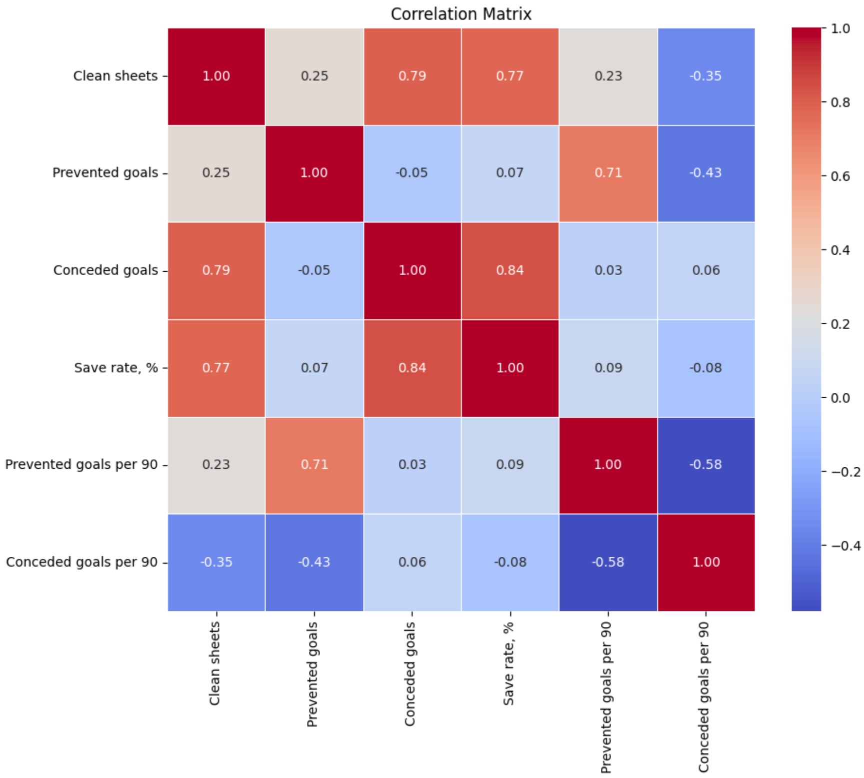

Figure 8 Correlation matrix of the variables

After undergoing EDA regarding my final model and business problem I found that the NAC Goalkeeper, compared to other keepers of similar level, was really lacking in terms of numbers and player statistics. What I did was take the NAC Keeper and others of the same league NAC play in, and compare them using the chosen features which include Prevented goals, clean sheets, etc. The results were underwhelming, and I did the same for goalkeepers that were of higher level, specifically from the Eredivisie, the highest Dutch professional football level and as expected the NAC keeper could not hold a candle to those others. Now knowing this, I need to take more care into the solution I bring to the problem since the situation is not the greatest, but I believe it can be solved. I also saw that average save rate % was higher amongst older players. The stat got higher as the age increased.

3 Machine Learning

3.1 Method:

The final machine learning model that I have chosen is the multiple linear regression model. Using multiple variables, you predict a continuous outcome variable based on the predictor variables, with the goal of minimizing the difference between the predicted values and the actual real values present in the dataset. It assumes linear relationships between the variables, and it represents relationships as straight lines. As far as machine learning models go, it is one of the simpler ones on the market, but some would call its simplicity one of the strong points of the linear regression. It is easy to implement and computationally does not take a lot.

The reason for the choosing of the linear regression model out of all the other ones is because in the context of my business problem I assumed that a linear combination of features would reasonably capture the relationship with the target, since the linear regression assumes a linear relationship between the input features and the target variable, giving it a more accurate score than other models would. Also, the linear regression performed well, so it was a no-brainer for me.

3.2 Model evaluations:

The methods I used to evaluate my final model were the Mean Squared Error (MSE), the Mean Absolute Error (MAE) and lastly the R-Squared (r2). When it comes to evaluating a linear regression model these are the key methods to evaluate the accuracy and efficacy of the model being evaluated. MAE measures the average absolute difference between the predicted and actual values. The MSE measures the the average squared difference between the predicted and actual values. And the r2 represents the proportion of the variance in the dependent variable that is predictable from the independent variables. The r2 is basically the accuracy score for the linear regression, it goes from 0 to 1. And as for MAE and MSE, the lower the score the better.

The scores my model received for MSE and MAE were quite good, as the scores I got were so low to the point that they were not able to show the full numbers on screen. As for the r2 I got a perfect 1.0, which is of course the highest you can get for the r2. It was so high that I questioned it and had worries of overfitting. So, I tested for that by doing a cross-validation using the neg_mean_squared_error scoring method as well as checking the training and testing set performance. The results showed that the model was not overfitted and luckily I got off free.

```
Mean Squared Error: 1.4187189657577117e-30
Mean Absolute Error: 3.5794126669939786e-16
R-squared: 1.0
```

Figure 9 Evaluation scores for my model

3.3 Model improvement:

I used hyperparameter tuning on one of the models from the template, I specifically used GridSeachCV (Cross-Validation) with the neg_mean_squared_error scoring method. It was to try and find the best set of combinations between the values to find the best performance, although I didn't use this extensively and it didn't carry any kind of importance in my final models. I also added a MinMax scaler to the model so that the ranges of the features used aren't influencing the score in a negative way and normalizes everything. But overall, the tuning of my model didn't play that big of a role in my final models.

4 Ethical Considerations

When working with data that can be considered personal data, data collected on individuals and not just a statistic with no name or face, which is the data we are working with, ethical considerations should be closely monitored and made sure to be followed. In terms of the organization, we are working with, NAC Breda, the three elements that they should follow are for it to be an ethical company, if the process and tools used are ethical and lastly to have ethical people (the employees and clients). The company itself seems to be ethical, following the GDPR guidelines. The data they provided is accurate and relevant I also assume that it is protected and the uses are explained in a transparent manner. For the processing of the data I would say NAC follows that too, if the data they gave us is in the clear, I also assume that the collection methods and processing methods for their data is ethical and up to standard. As for their staff, we've met one of the scouts and one of the people from the data department and they seem like good people, but one issue I found is that looking through their staff, most of them are Dutch. This includes the coaching staff, the scouts, the management and the board of directors, all of them are Dutch. I would recommend including some diversity into the staff as that follows ethical guidelines. It promotes equal opportunity and avoids any discrimination. Another thing I need to mention is when one of the members of the scouring team visited the University, he mentioned that only players from Europe are accepted into the team mostly for economic reasons. This falls into the same ethical problems I just discussed. If it is possible to increase the budget for scouting and recruiting foreign players, then there can be more diversity within the squad and the team might also find some hidden talents that wouldn't have appeared before.

As for me, I used the data provided in fair and transparent manner. I do not have any control or ownership over this data, and I assume that the clients already know what is being done with their data. I did not share this data with anyone else that is outside of the scope of this project meaning that the data is fully protected and everything that I have done has been explained with full clarity, leaving nothing out of it. I simply used and analysed the data for the sake of the client and the problem statement and to ultimately help the client in whatever way I can. With that I believe I have followed the GDPR guidelines.

5 Recommendations

I will now go back to my main and final model and explain a few extra EDA findings related to the problem statement and final model. As I have said at the start of this report, I want to find a way to improve the NAC Breda goalkeeper by figuring out the best statistics to improve on. To do that I have created custom score called the GK Score. The way this works is that I combined several different statistics together using weighted scoring. Those statistics include: Clean sheets, prevented goals, conceded goals, save rate %, prevented goals per 90 and conceded goals per 90. The score ranges from 0 to 1, 0 being bad and 1 meaning good. I fine-tuned it to make it just for Goalkeepers because of the used stats. I assign weights to the different stats, that dictates the importance of each stat when calculating the Custom score, and I assigned equal importance across all of the metrics. For conceded goals and conceded goals per 90, I have assigned the weights to be negative. The reason for that is if a goalkeeper has an increase in conceded goals, then as a result the overall score should lower itself since that is something negative in football and something to be avoided. So how that works, if a goalkeeper has less conceded goals, then their score will go up accordingly.

Then I compared other Goalkeepers of similar level to the NAC Keeper using the GK Score. The NAC Keeper came in at almost last in terms of score, highlighting the importance of him having to improve.

Player	Gk Score
L. Koopmans	**0.72**
Joao Virginia	**0.66**
J. Smiths	**0.57**
R, Ruiter	**0.45**
R. Kortsmit (NAC)	**0.40**
M. Verrips	**0.37**

Figure 9 Table of GK Scores

Then using the GK Score I ran that with the selected features in a linear regression model and calculated/predicted the stats that have the most impact on the GK Score. The results are as follows: For clean sheets, an increase of one-standard-deviation would yield an increase of 1.56% on the GK Score. Now, how much is a one-standard-deviation? We can find that out by adding the standard deviation and the mean of the 'Clean sheets' column we get 1.49. What this means is that when there is an increase of roughly 1.5 (1.49) clean sheets then the GK Score will go up 1.56%. And when there is an increase in conceded goals, there will be a decrease of 2.1 % in the GK Score.

Statistic	Percent increase/decrease
Save rate %	**3.8%**
Cleans sheets	**1.56%**
Prevented goals	**0.79%**
Prevented goals per 90	**0.38%**
Conceded goals per 90	**-0.57%**
Conceded goals	**-2.15%**

Figure 10 Table of percentage increases/decreases

This has illustrated just what exact stats to focus on for the NAC Keeper to get better. There is a clear need for the NAC Goalkeeper to get better at his game as we saw with the comparisons. If the NAC keeper improves in these statistics, their GK Score will increase, making their performance increase as well, which therefore makes their Market value increase, making it a way to predict market value.

This was just the most basic version of this and the first time I am experimenting with anything of this kind. This can also be applied to other positions and can be performed at a larger scale. The final recommendations that I can make based on the results are to really focus on Save rate % and conceded goals, Of course, every stat is important, and everything will contribute to a better score, but it is also important to highlight the most important parts and give focus to those parts like we have done here.

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